

MBZUAI-Focused ML Portfolio Roadmap (4 Months)

Objective

Build a small but technically rigorous ML portfolio that demonstrates:

- Strong ML engineering fundamentals
- Research-oriented thinking
- Ability to implement and evaluate models independently
- Understanding of data pipelines, experimentation, and deployment
- Evidence of mathematical maturity and reproducibility

The goal is not to build many beginner projects. The goal is to produce 3–4 high-quality projects with:

- Clean GitHub repositories
- Strong READMEs
- Proper experimentation
- Evaluation metrics
- Reports/blogs
- Optional deployment

For MBZUAI applications, depth matters more than quantity.

Recommended Project Stack

You should aim for the following portfolio structure:

Project Type	Difficulty	Admission Value
End-to-End Classical ML Project	Medium	Shows engineering maturity
Deep Learning Project	Medium-High	Shows modeling capability
Research-Style Reproduction Project	High	Very valuable for MBZUAI
Optional MLOps/Deployment Project	Medium	Shows practical engineering

Core Technology Stack

Programming

- Python
- SQL
- Bash basics

ML Libraries

- NumPy
- Pandas
- Scikit-learn
- XGBoost
- LightGBM
- PyTorch
- Hugging Face Transformers

Visualization

- Matplotlib
- Seaborn
- Plotly

MLOps / Engineering

- Git + GitHub
- MLflow
- Docker
- FastAPI
- Streamlit
- DVC (optional but valuable)

Cloud (Optional)

- Google Colab
 - Kaggle
 - AWS basics
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Four-Month Roadmap

Month 1 — Build Strong ML Foundations + Upgrade Existing Project

Goal

Transform your Student Performance project into a professional-grade repository.

Most applicants stop after model training. You must go beyond that.

Week 1 — Rebuild Student Performance Project Properly

Tasks

- Reorganize project structure
- Add modular pipeline architecture
- Add:
 - data ingestion
 - preprocessing
 - training pipeline
 - evaluation pipeline
 - prediction pipeline

Recommended Structure

```
project/  
├── data/  
├── notebooks/  
├── src/  
│   ├── components/  
│   ├── pipelines/  
│   ├── utils/  
│   └── exception.py  
├── app.py  
├── requirements.txt  
├── setup.py  
├── README.md  
└── experiments/
```

Add These Features

- Cross-validation
- Hyperparameter tuning
- Feature importance analysis
- Error analysis
- Experiment tracking

Models

- Linear Regression
- Random Forest
- XGBoost
- CatBoost

Week 2 — Deployment + Documentation

Tasks

- Build Streamlit or FastAPI app
- Dockerize project
- Deploy online

Deliverables

- Live demo
- GitHub repo
- Professional README
- Architecture diagram

README Must Include

- Problem statement
 - Dataset
 - Pipeline architecture
 - Models tried
 - Metrics comparison
 - Deployment screenshots
 - Future improvements
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Week 3 — Advanced ML Concepts

Learn Properly

- Bias vs variance
- Regularization
- Feature engineering
- Ensemble learning
- Gradient boosting
- Evaluation metrics

Implement From Scratch

This matters for research-oriented admissions.

Implement from scratch using NumPy:

- Linear Regression
- Logistic Regression
- Gradient Descent
- K-Means

Do NOT copy code blindly.

Week 4 — Research Reading Habit

Start reading papers.

Read:

- Attention Is All You Need
- BERT
- ResNet
- XGBoost paper

Goal

You do NOT need to understand every equation.

You should learn:

- Problem formulation
- Methodology
- Experimental setup
- Evaluation strategy

Output

Create:

- 1-page summaries
- LinkedIn posts
- GitHub notes repository

This signals intellectual curiosity.

Month 2 — Deep Learning Project

Goal

Build one serious deep learning project.

Choose ONE domain:

- NLP
- Computer Vision
- Time Series

For MBZUAI, NLP is especially valuable.

Recommended Project 1 (Strongly Recommended)

Fake News Detection Using Transformers

Why This Project?

It demonstrates:

- NLP understanding
- Transformer usage
- Fine-tuning
- Modern DL workflow

Stack

- PyTorch
- Hugging Face
- BERT or DistilBERT

Features

- Data preprocessing
- Tokenization
- Fine-tuning
- Metrics dashboard
- Error analysis

Advanced Additions

- Compare BERT vs DistilBERT
- Add explainability using SHAP
- Build inference API
- Add confidence scoring

Alternative Projects

Computer Vision

- Pneumonia Detection
- Plant Disease Classification
- Road Damage Detection

Time Series

- Electricity Demand Forecasting
- Crypto Price Forecasting
- Air Quality Prediction

Month 2 Weekly Plan

Week 1

- Learn PyTorch fundamentals
- Tensors
- Autograd
- Training loops
- Dataloaders

Week 2

- Build baseline neural network
- Train and evaluate

Week 3

- Fine-tune pretrained model
- Experiment tracking
- Hyperparameter tuning

Week 4

- Deployment
- Documentation
- Technical blog

Month 3 — Research-Oriented Project (HIGH VALUE)

Goal

This is the project that can differentiate your profile.

You will reproduce or improve a research paper.

This is extremely valuable for MBZUAI.

Recommended Research-Style Project

Paper Reproduction Project

Choose One

- BERT fine-tuning paper
 - Vision Transformer
 - CLIP
 - LoRA fine-tuning
 - Retrieval-Augmented Generation (RAG)
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Recommended Option

Retrieval-Augmented QA System

Why?

RAG is highly relevant in modern AI.

Skills Demonstrated

- Embeddings
- Vector databases
- LLM pipelines
- Retrieval systems
- Evaluation methodology

Stack

- LangChain
- FAISS
- Sentence Transformers
- Open-source LLM

Features

- Document ingestion
- Chunking
- Embedding pipeline
- Semantic search
- Question answering
- Evaluation metrics

Advanced Features

- Hybrid retrieval
- Reranking
- Memory system

- Hallucination evaluation
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Month 3 Weekly Plan

Week 1

- Read paper carefully
- Understand architecture
- Reproduce small components

Week 2

- Build full baseline implementation

Week 3

- Improve one aspect Examples:
 - better preprocessing
 - different embeddings
 - better retriever
 - improved evaluation

Week 4

- Write technical report
 - Create architecture diagrams
 - Prepare portfolio presentation
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Month 4 — Portfolio Polishing + Interview Preparation

Goal

Convert projects into admission-ready assets.

Week 1 — GitHub Optimization

Every repository should contain:

- Proper README
- Setup instructions
- Results
- Screenshots

- Architecture diagrams
- Experiment details
- Future work

Add

- Requirements file
 - Docker support
 - CI/CD basics (optional)
 - Unit tests (basic)
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Week 2 — Technical Blogging

Write 3–5 blogs:

Suggested Topics

- How I built a transformer classifier
- Understanding gradient boosting
- What I learned reproducing a paper
- RAG systems explained simply

Publish on:

- Medium
 - Hashnode
 - LinkedIn
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Week 3 — Mathematics Revision

MBZUAI values strong math.

Focus on:

- Linear algebra
- Probability
- Statistics
- Calculus for ML
- Optimization

Important Topics

Linear Algebra

- Eigenvalues
- Eigenvectors
- Matrix multiplication
- SVD

Statistics

- Probability distributions
- Bayes theorem
- Hypothesis testing
- Expectation and variance

Calculus

- Derivatives
 - Chain rule
 - Gradient descent
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Week 4 — Interview + SOP Preparation

Prepare to Explain:

- Why you chose each model
- Tradeoffs
- Failure cases
- Dataset limitations
- Evaluation metrics
- Overfitting prevention
- Deployment decisions

You Must Be Able To Explain

- Bias-variance tradeoff
 - Gradient descent
 - Attention mechanism
 - Transformer architecture
 - Cross-validation
 - Precision vs recall
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Final Portfolio Recommendation

By the end of 4 months, your profile should contain:

Project 1

Student Performance ML System

Features:

- End-to-end pipeline
- Deployment
- Docker
- Experiment tracking

Project 2

Transformer-Based NLP Project

Features:

- PyTorch
- Hugging Face
- Fine-tuning
- Explainability

Project 3

Research Reproduction / RAG Project

Features:

- Paper implementation
- Technical report
- Experimental comparison
- Research thinking

Supporting Assets

- Technical blogs
- Clean GitHub
- Mathematics notes
- Paper summaries
- LinkedIn activity

What MBZUAI Admissions Usually Value

High Value Signals

- Research potential
- Strong mathematics
- Independent implementation
- Reproducibility
- Deep learning understanding
- Technical communication

Low Value Signals

- Tutorial-copy projects

- Weak documentation
 - Only Kaggle notebooks
 - No experimentation
 - No deployment
 - No explanation of decisions
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Recommended Daily Schedule

Weekdays

Activity	Duration
Mathematics	1 hour
ML Theory	1 hour
Coding/Project	2–3 hours
Paper Reading	30 mins

Recommended Resources

Mathematics

- Linear Algebra by Gilbert Strang
- 3Blue1Brown
- StatQuest

ML Theory

- Hands-On Machine Learning
- CS229
- Andrew Ng ML Specialization

Deep Learning

- Deep Learning Specialization
- FastAI
- Stanford CS231n
- Stanford CS224n

Research

- Papers With Code
 - arXiv
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Important Strategic Advice

Do not build 10 beginner projects.

Three strong projects with:

- rigorous experimentation
- clean engineering
- technical depth
- strong explanations

are significantly more valuable.

The strongest differentiator for you will likely be: 1. Strong GitHub engineering 2. Research-style implementation 3. Ability to explain decisions deeply 4. Mathematical clarity

If you want, the next step can be:

- a project-by-project execution checklist,
- a GitHub portfolio structure,
- a daily study timetable,
- MBZUAI SOP strategy,
- or a list of research-grade project ideas specifically aligned with NLP, CV, or Generative AI.